HAT-P-19b

R-1088

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-19_241121` · created 2026-07-30T16:42:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460635.6783 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.016
Transit depth (Rp/Rs)^2	2.17 +/- 0.46 [%]
Orbital Inclination (inc)	88.29 +/- 0.58
Airmass coefficient 1 (a1)	0.999 +/- 0.011
Airmass coefficient 2 (a2)	-0.001 +/- 0.043
Scatter in the residuals of the lightcurve fit is	1.1 %
Best Comparison Star	#1 - [491, 312]
Optimal Aperture	3.25
Optimal Annulus	11.33
Transit Duration (day)	0.1099 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.147 ± 0.016 vs 0.1418 (deviation 0.31σ)
Duration fit vs published	0.1099 d vs 0.1196 d (deviation 1.94σ)
Mid-time O–C	-5.9 min
Depth SNR	8.8
Residual scatter	1.1 %

Light curve

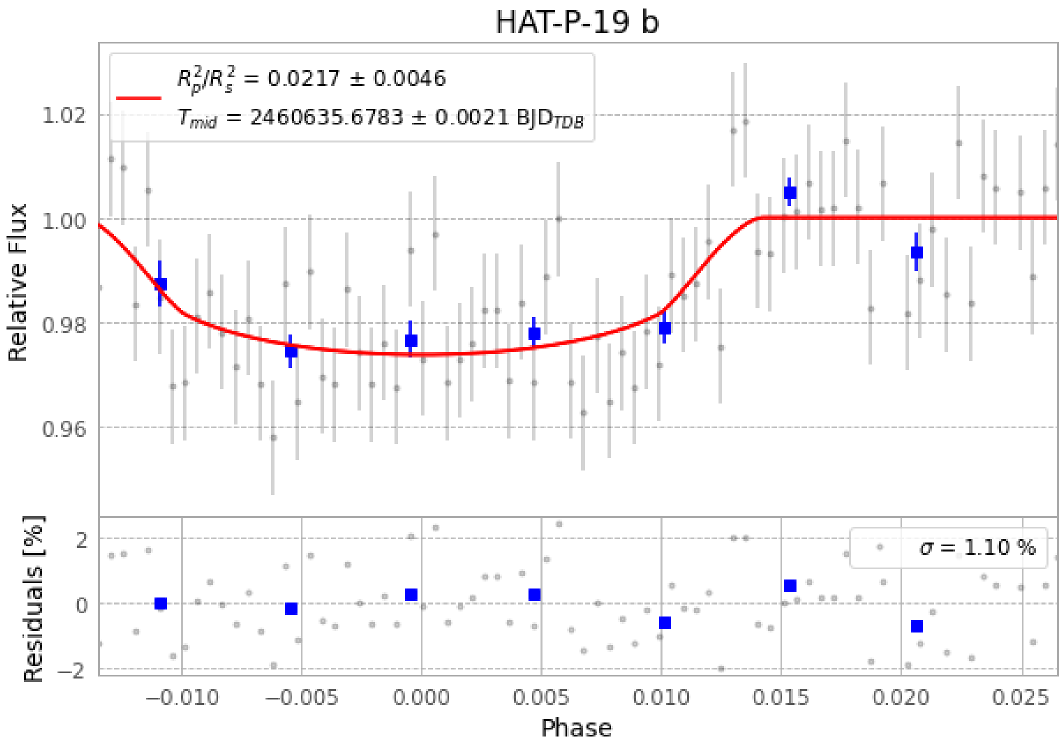

Figure 1 — FinalLightCurve_HAT-P-19 b_2024-11-20.png (SHA-256 b99b385cacb1aef7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [404, 250], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ab494804c99fc9eb…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

78 FITS frames (51 MB), `HATP-19241121025117.FITS` → `HATP-19241121064216.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-241121.log (09b9ed68572f…), FinalLightCurve_HAT-P-19 b_2024-11-20.png (b99b385cacb1…), FinalLightCurve_HAT-P-19 b_2024-11-20.csv (77bafe82f970…)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 16:42Z.